Allen Dufort

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Education

Brown University — *Providence, RI*
Master of Science in Computer Science*, GPA: 4.0* — **May 2025**
Bachelor of Science in Computer Science*, GPA: 3.85* — **May 2024**

Technical Skills

Languages: Python, SQL, Golang, JavaScript, Java, C++, C, HTML, CSS
Tools & Frameworks: watsonx.ai, watsonx Orchestrate, IBM Maximo, Hugging Face, OpenAI API, Apache Spark, Terraform, MongoDB, AWS, Azure, Git, WordPress
Areas of Expertise: Generative AI, Large Language Models, Agentic & Multi-Agent Systems, Machine Learning, Deep Learning, Predictive Modeling, NLP, Computer Vision, Data Analysis & Visualization, Responsible AI & AI Ethics, Quality Assurance Testing, UI/UX
Certifications: IBM AI Associate Data Scientist, watsonx.ai Generative AI Models Technical Sales Intermediate, Data Science Tools, Data Science Methodologies, IBM Cloud Essentials, IBM Garage Essentials, IBM Agile Explorer, Enterprise Design Thinking Practitioner
Spoken Languages: English (fluent), French (proficient), Spanish (proficient)

Experience

Associate Data Scientist, *IBM* — *Chicago, IL* | **2025 – Present**

- Built and delivered a generative AI supervisor assistant in watsonx Orchestrate that centralizes work-order information, flags anomalies, and streamlines review and approval workflows for a global pharmaceutical enterprise
- Reduced work-order review time by 75 percent across 300+ work orders per week, unlocking roughly 10.5K labor hours of capacity and $1.12M in repurposed labor value annually
- Translated business use cases into AI solution architecture with clients, and produced documentation and demos that enabled adoption across delivery teams
- Developed Python-based AI compliance tools and a custom API wrapper integrating live enterprise asset-management (Maximo) data, with resilient handling of upstream failures
- Improved AI scalability, throughput, and latency through optimized model inference, efficient tool orchestration, parallel processing, and robust error handling; built a unit testing framework and ran integration testing to prevent functional and AI accuracy defects
- Developed AI/ML models and dashboards for an energy enterprise to proactively predict and prevent serious injuries and fatalities, automating content summarization, classification, and compliance checking while partnering with subject-matter experts to validate model outputs
- Integrated 2024 release core code into a semiconductor manufacturer's custom C++ codebase for IBM's SiView Manufacturing Execution System, independently completing and unit testing merges across three subsystems (specification management, material transport, and floor control)

Graduate Teaching Assistant, *Brown University* — *Providence, RI* | **January 2025 – May 2025**

- Created course content and led live demos teaching students to design multi-agent systems with generative AI and build large language model applications using Python, Jupyter Notebook, and the OpenAI API
- Graded assignments and held office hours, guiding students in debugging code, optimizing algorithms, and applying generative AI techniques to practical use cases

Software Engineering Intern, *Marqeta* — *Remote* | **June 2024 – August 2024**

- Developed a GitHub application integrated with the CI team's service, issuing on-demand personal access tokens scoped to user-specified repositories; the design was adopted as a team best practice
- Reduced token retrieval time by 95 percent and delivered documentation and demos that drove company-wide adoption, enhancing the CI/CD pipeline's efficiency

Software Engineering Intern, *General Motors* — *Roswell, GA* | **June 2023 – August 2023**

- Implemented front-end features for Account Overview used by over 22 million online users
- Led testing phase, automating and executing tests to identify critical bugs and enhance product quality
- Engaged in scrum practices, including creating and reviewing PRs, bug triages, PI planning, and deployments

Software Engineering Research Assistant, *Brown University Physics Department* — *Providence, RI* | **November 2021 – May 20**

- Owned backend software for the cryostat control system of NASA's Exoplanet Climate Infrared Telescope (EXCITE), managing commands and data exchange between the ground computer and temperature-sensitive sensors
- Diagnosed and resolved cross-system communication issues via robust serial and network connections, and collaborated with NASA's Goddard Space Flight Center and partner institutions to plan, test, and document solutions

Freelance Web Developer, *Diabetic and Hypertension Relief Center* — *Remote* | **January 2015 – Present**

- Oversee UI design, including CSS/HTML elements, text updates, and photo uploads, for the WordPress-supported site dhreliefcenter.org since 2021

Projects

Language Models as Reward Functions for RL | *Python, Jupyter Notebook, Gymnasium, Hugging Face* **Spring 2025**

- Developed a reinforcement learning framework using large language models as reward signal generators instead of traditional mathematical reward functions
- Implemented and evaluated Q-learning agents in Frozen Lake and Blackjack environments, comparing LLM-based rewards against numeric baselines across 400–500 training episodes, and showed language-driven rewards yield more flexible, human-aligned feedback

Iterative Design | *HTML, CSS, Figma, Balsamiq* **Spring 2025**

- Redesigned a healthcare platform to improve accessibility and personalization for AI-driven healthcare tools, developing iterative prototypes refined through structured critique cycles with industry executives

JeopardyAI! | *Python, Jupyter Notebook, AutoGen, OpenAI* **Fall 2024**

- Built a Jeopardy-style trivia game with a multi-agent system that dynamically generated questions, moderated gameplay, and kept score against an AI game host
- Integrated OpenAI and Cluebase APIs to deliver an engaging experience with access to over 350,000 trivia questions

Honors and Awards

Brown University Computer Science Scholarship **March 2024**

- Merit-based full-tuition scholarship for the Master's Program of Computer Science at Brown

Brown University Undergraduate Teaching and Research Award **Spring 2024, Spring 2023, Fall 2022**

- Supports Brown students collaborating with Brown faculty on research

NASA Rhode Island Space Grant Summer Scholarship **April 2022**

- Awarded to undergraduates carrying out NASA-related research projects

Dr. Martin Luther King Jr. Trust Fund Scholarship **October 2019**

- Awarded to college-bound students with exceptional academics and community service experience